TransitDuet: Bi-Level Reinforcement Learning with Temporal Advantage Propagation for Joint Bus Timetable Planning and Holding Control

Anonymous Authors[a]

[a] *Anonymous Institution*

Abstract

We present TransitDuet, a bi-level reinforcement learning framework that jointly optimizes bus timetable planning and real-time holding control. This joint problem is fundamentally harder than either task in isolation: the planning and control layers operate on different timescales, observe different state spaces, and interact through delayed, asynchronous feedback—properties that violate the synchronous co-execution assumption underlying existing hierarchical reinforcement learning methods. TransitDuet addresses this through three mechanisms. First, Temporal Advantage Propagation aggregates cross-level rewards over complete trip lifecycles to couple an upper dispatch policy with a lower holding policy, generalizing synchronous cross-advantage to asynchronous multi-timescale settings. Second, a measurement-projection scheme based on online gradient descent enforces fleet-size constraints without reward shaping. Third, Lagrangian relaxation within the lower policy penalizes headway deviations, converting a hard service-regularity requirement into a soft dual objective. We evaluate TransitDuet in a calibrated microsimulation of a high-frequency bus corridor with stochastic demand and traffic. Compared to fixed-timetable holding control

and single-level planning baselines, TransitDuet reduces passenger waiting time by **[TBD]**% while satisfying fleet and headway constraints, demonstrating that tightly coupling planning and control through asynchronous advantage propagation yields benefits unattainable by either layer alone.
Keywords: Bus holding control, Timetable optimization, Hierarchical reinforcement learning, Multi-timescale control, Constrained optimization

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1. Introduction

High-frequency bus systems face a persistent coordination failure. Timetables are planned offline to balance fleet utilization against passenger demand, yet the headways they prescribe degrade within minutes once buses encounter stochastic traffic and boarding delays. Real-time holding control can restore regularity, but only within the headway structure it inherits. When the timetable itself is poorly matched to demand—dispatching too few buses in the peak or too many in the off-peak—no amount of holding can recover acceptable service. Planning and control are thus coupled: the quality of one layer bounds the achievable performance of the other.

Despite this coupling, existing work treats the two problems separately. The timetable optimization literature formulates headway or frequency design as mixed-integer programs solved offline [1], while the holding control literature trains reinforcement learning (RL) agents that take the timetable as given [2]. Joint optimization would seem a natural application of hierarchical RL, yet three gaps prevent a straightforward application.

Asynchrony between planning and control.. Hierarchical RL frameworks such as the options framework [3] and goal-conditioned methods like HIRO [4]

assume that the upper and lower levels share a common state clock or that the upper level sets subgoals consumed by the lower level within a fixed horizon. In bus operations, the upper policy acts at dispatch events (once per bus departure), while the lower policy acts at every station arrival across all active buses. The two levels observe different state spaces—the upper level sees route-level demand forecasts and fleet status, whereas the lower level sees per-bus headway deviations and passenger loads—and interact through delayed feedback spanning an entire trip lifecycle. No shared clock or fixed subgoal horizon exists.

Cross-advantage under asynchrony.. **(author?)** [5] recently introduced cross-advantage augmentation for bi-level robot locomotion, where an upper style policy and a lower tracking policy execute synchronously on the same state. Their augmented advantage $\tilde{A}_{\mathrm{U}} = A_{\mathrm{U}} + \beta A_{\mathrm{L}}$ directly sums the two advantages at each time step. This elegant mechanism breaks down when the two levels operate on different timescales with different state and action spaces: there is no single time step at which A_{U} and A_{L} are jointly defined. Generalizing cross-advantage to the asynchronous, multi-timescale setting requires aggregating lower-level feedback over variable-length intervals aligned with upper-level decisions.

Fleet and regularity constraints.. Bus operations impose hard constraints—a fixed fleet size N_{fleet} and bounded headway deviation—that reward shaping handles poorly. Constrained RL methods such as RCPO [6] address single-level constraint satisfaction but do not account for constraints that span a bi-level structure, where the upper level controls resource allocation and the lower level controls service regularity.

We propose **TransitDuet**, a bi-level RL framework that addresses all three gaps. The upper policy π_{U} outputs target headways $[H_{\mathrm{peak}}, H_{\mathrm{off}}, H_{\mathrm{trans}}]$ at each dispatch event, while the lower policy π_{L}, a distributional soft actor-critic with Lagrangian penalties [7], selects holding times at every station arrival with parameter sharing across buses. The two levels are coupled through *Temporal Advantage Propagation* (TAP): the upper advantage is augmented with discounted lower-level rewards accumulated over the trip lifecycle triggered by each dispatch, and vice versa. A cosine-annealed coupling coefficient β ramps from zero to $\beta_{\max}$ after an initial warmup phase, preventing premature coupling before either policy has learned a reasonable baseline. Fleet-size feasibility is enforced by a $\boldsymbol{\theta}$-weighted online gradient descent projection on the measurement vector $\mathbf{z}$ of dispatched buses, inspired by the ApproPO framework [8], while headway uniformity enters the lower-level objective through Lagrangian relaxation.

Contributions.. The specific contributions of this paper are:

1. **Temporal Advantage Propagation.** We formalize TAP, an asynchronous cross-advantage mechanism that aggregates rewards across timescales via trip-lifecycle alignment. We prove that TAP preserves policy-gradient unbiasedness under standard assumptions and reduces to the synchronous cross-advantage of **(author?)** [5] when both levels share a common clock.

2. **Joint timetable–holding formulation.** We cast bus timetable planning and holding control as a bi-level Markov decision process with heterogeneous state-action spaces and asynchronous transitions, providing

a formal problem structure absent from prior bus RL work.

3. **Constraint satisfaction via $\boldsymbol{\theta}$-OGD and Lagrangian relaxation.** We integrate measurement-projection constraints at the upper level and dual-variable penalties at the lower level, handling fleet feasibility and headway regularity without reward shaping.

4. **Empirical evaluation on a calibrated corridor.** We demonstrate on a stochastic microsimulation that TransitDuet reduces passenger waiting time by **[TBD]**% over the best single-level baseline while maintaining fleet and headway constraint satisfaction, and ablations confirm that each component—TAP, β annealing, $\boldsymbol{\theta}$-OGD—is necessary for this result.

Results preview.. In experiments on a high-frequency bus corridor with **[TBD]** stops and calibrated demand profiles, TransitDuet achieves a **[TBD]**% reduction in mean passenger waiting time and a **[TBD]**% reduction in headway coefficient of variation compared to a fixed-timetable DSAC holding controller, while satisfying the fleet constraint in **[TBD]**% of episodes.

Paper organization.. Section 2 reviews related work in timetable optimization, bus holding RL, and hierarchical RL. Section 3 formalizes the bi-level MDP. Section 4 presents the TransitDuet architecture, TAP, and constraint-handling mechanisms. Section 5 describes the simulation environment and reports results. Section 6 discusses limitations and extensions. Section 7 concludes.

2. Related Work

2.1. Reinforcement Learning for Bus Holding Control

The application of reinforcement learning to real-time bus holding control has attracted considerable attention in recent years, driven by the promise of adaptive policies that can respond to stochastic passenger demand and traffic conditions. Early work by **(author?)** [2] formulated the holding problem as a Markov decision process and applied deep Q-networks (DQN) to learn stop-level holding durations, demonstrating that learned policies could reduce headway variance compared to rule-based strategies. **(author?)** [9] extended this line of work with policy-gradient methods, showing that PPO-based agents achieve more stable training and better generalization across demand scenarios than value-based alternatives. More recently, **(author?)** [10] investigated soft actor-critic (SAC) for holding control and introduced a parameter-sharing architecture in which a single policy network is deployed across all buses on a route, with bus-specific observations enabling heterogeneous behavior from a shared set of parameters. This parameter-sharing paradigm has become the de facto approach for multi-bus control, as it scales gracefully with fleet size and implicitly encourages cooperative behavior through shared gradient updates.

Despite these advances, existing RL-based holding controllers share a fundamental limitation: they operate under a fixed, externally supplied timetable. The planned departure times and target headways are treated as immutable inputs, meaning that the holding policy has no mechanism to communicate execution-level feedback—such as systematic schedule infeasibility or recurrent bunching patterns—back to the planning stage. When

the timetable itself is poorly calibrated to actual operating conditions, even an optimal holding policy can only mitigate, not resolve, the resulting service irregularity. TransitDuet addresses this gap by embedding the holding controller as the low-level agent within a bi-level framework, where real-time execution performance directly shapes the timetable through Temporal Advantage Propagation.

2.2. Bus Timetable Optimization

Timetable design has traditionally been approached through mathematical programming and metaheuristic search. **(author?)** [11] formulated the problem as a mixed-integer program that jointly determines departure times and vehicle assignments to minimize a weighted combination of passenger waiting time and operator cost, solving it via branch-and-bound with valid inequalities. **(author?)** [1] proposed a bi-level model in which the upper level selects headways to minimize total passenger cost while the lower level solves a vehicle scheduling problem, solved using an adaptive large neighborhood search. Frequency-setting variants, such as the work of **(author?)** [12], optimize the number of departures per period rather than individual departure times, and typically rely on assignment models to estimate passenger flows under each candidate frequency vector.

A common thread in this literature is the reliance on static demand forecasts and deterministic travel-time assumptions. Timetables are optimized offline against expected conditions and then handed off to operations without a feedback loop. When actual conditions deviate—due to traffic incidents, demand surges, or driver variability—the timetable may become infeasible, and holding controllers are left to absorb the mismatch. Moreover, the com-

binatorial structure of these formulations limits scalability: solution times grow rapidly with the number of trips and stops, making re-optimization during the operating day impractical. TransitDuet replaces this open-loop paradigm with a closed-loop one. The high-level agent learns a timetable policy that maps aggregate state features to departure-time adjustments, and it receives gradient signals from low-level execution via TAP, enabling the planner to adapt its decisions based on how well the resulting schedule can actually be executed.

2.3. Hierarchical and Multi-Timescale Reinforcement Learning

Hierarchical reinforcement learning decomposes complex tasks into temporally extended sub-problems. The options framework of **(author?)** [3] introduced the notion of temporally abstract actions, each consisting of an initiation set, an intra-option policy, and a termination condition, enabling planning over multiple timescales within a single MDP. Feudal networks [13] operationalize this idea with a manager that sets sub-goals in a learned latent space and a worker that executes primitive actions to reach them, training both levels end-to-end with the manager's policy gradient modulated by the worker's goal-reaching success. HIRO [4] further improved sample efficiency by introducing off-policy corrections that relabel high-level goals in hindsight, making hierarchical training practical in continuous-control domains. Beyond goal-conditioned hierarchies, **(author?)** [14] and **(author?)** [15] studied the multi-timescale setting more broadly, analyzing the interaction between agents operating at different decision frequencies and establishing regret bounds for the resulting non-stationary sub-problems.

While these methods provide powerful abstractions for temporal hierar-

chy, most assume either synchronous execution—where the high-level decision triggers immediately before a fixed-length sequence of low-level steps—or a shared state space across levels. In transit operations, the planning and control timescales are fundamentally asynchronous: the timetable agent acts once per planning horizon (e.g., every 15–30 minutes), while the holding agent acts at every stop visit (every 1–3 minutes), and the two levels observe different state representations at non-aligned decision epochs. TransitDuet introduces Temporal Advantage Propagation to handle this asynchrony. TAP generalizes the cross-advantage mechanism of RoboDuet from synchronized paired decisions to temporally misaligned ones, accumulating discounted low-level advantages over the variable-length interval between consecutive high-level actions and injecting the resulting signal into the high-level policy gradient. This allows the planner to receive credit assignment from execution outcomes without requiring the two levels to share a clock or a state space.

2.4. Constrained Reinforcement Learning

Many practical RL deployments must satisfy constraints beyond reward maximization. Lagrangian relaxation is the most widely used approach: **(author?)** [6] proposed reward-constrained policy optimization, which augments the policy gradient with dual variables corresponding to cost constraints and performs primal-dual updates to steer the policy toward feasibility. **(author?)** [16] improved upon this with PID-based Lagrange multiplier updates that reduce oscillation and speed convergence in the constrained setting. These methods are effective for soft constraints, where occasional small violations are tolerable, but they offer no worst-case guarantees—the Lagrange multiplier may fail to grow fast enough to prevent persistent violations during

training.

For settings that demand hard constraint satisfaction, **(author?)** [8] introduced ApproPO, which formulates the constrained problem as an online convex optimization over the space of occupancy measures and applies a measurement-based projection oracle to enforce feasibility at every iterate. The θ-OGD variant provides a practical implementation that projects policy parameters onto the feasible set using online gradient descent with approximate projections. In the multi-agent domain, **(author?)** [5] proposed RoboDuet for humanoid locomotion, where two coupled agents—an upper-body controller and a lower-body locomotor—exchange cross-advantage terms that capture how each agent's action quality is perceived under the other's value function, enabling tight coordination without centralized control.

TransitDuet combines elements from both lines of work to address the distinct constraint structure of transit operations. Fleet-size and capacity constraints are hard operational requirements that must never be violated; we enforce these through ApproPO θ-OGD applied to the high-level timetable policy, guaranteeing that planned departures respect vehicle availability at every training iterate. Headway uniformity, by contrast, is a softer service-quality objective that admits graceful degradation; we handle it via a Lagrangian penalty with adaptive multiplier updates in the low-level holding policy. The cross-level coordination itself is managed by TAP, which extends RoboDuet's synchronous cross-advantage to the asynchronous, multi-timescale regime that characterizes the planning–control interface. This layered constraint architecture allows TransitDuet to provide hard guarantees where they are operationally necessary while retaining the flexibility of soft

penalties where service trade-offs are acceptable.

3. Problem Formulation

We formalize joint timetable planning and holding control as a bi-level Markov decision process (MDP) operating over a single bus route with time-varying passenger demand. This section first describes the transit system model (Section 3.1), then defines the upper- and lower-level MDPs (Section 3.2), states the constraint specification (Section 3.3), and concludes with a discussion of the timescale separation that motivates our algorithmic design (Section 3.4).

3.1. Bus Transit System Model

We consider a single bidirectional bus route consisting of M stations indexed $\{1, \ldots, M\}$. Service is provided according to a timetable that dispatches a total of 264 trips per day, alternating between the two directions. Passenger demand is drawn from time-varying origin–destination (OD) matrices that capture morning, midday, and evening peaks. At any point in time, at most $N_{\text{fleet}}^{\max} = 25$ buses may be concurrently in service; each trip is characterized by a launch time and a travel direction. The dispatch headway—the time between consecutive launches in the same direction—is the primary planning lever, while per-station holding times constitute the real-time control lever.

3.2. Bi-Level MDP Formulation

We decompose the joint optimization into two coupled MDPs that share a common environment but differ in observation granularity, action semantics,

and decision frequency.

3.2.1. Upper-Level MDP (Timetable Planning)

The upper-level policy π_{U} determines dispatch headways. It is triggered at each dispatch event, yielding approximately 132 decision epochs per episode (one full service day).

State.. The upper state encodes system-level summary statistics:

$$s_{\mathrm{U}} = \left[\frac{\mathrm{hour}}{24},\ \frac{\mathrm{demand}}{1000},\ \frac{\mathrm{fleet_on_route}}{N_{\mathrm{fleet}}^{\max}},\ \frac{\mathrm{prev_headway}}{600},\ \mathrm{unhealthy_rate} \right] \in \mathbb{R}^5, \quad (1)$$

where each component is normalized to facilitate learning. The *unhealthy_rate* captures the fraction of active buses whose observed headway deviates from the target by more than a tolerance threshold.

Action.. The upper action specifies headways for three demand regimes:

$$a_{\mathrm{U}} = \left[H_{\mathrm{peak}},\ H_{\mathrm{off}},\ H_{\mathrm{trans}} \right] \in \mathbb{R}^3, \quad (2)$$

with ranges $H_{\mathrm{peak}} \in [180, 600]$, $H_{\mathrm{off}} \in [300, 1200]$, and $H_{\mathrm{trans}} \in [240, 900]$, all in seconds. These three headways correspond to peak, off-peak, and transitional periods, respectively. The environment selects the applicable headway based on the current time of day and uses it as both the dispatch interval and the target headway h^{target} communicated to the lower level.

Reward.. The upper reward is computed at the episode level from a measurement vector (defined in Section 3.3). Because each upper action influences the system over an extended horizon through the downstream holding decisions it induces, assigning credit to individual dispatch events is nontrivial; we defer the reward construction to Section 4.

3.2.2. Lower-Level MDP (Holding Control)

The lower-level policy π_{L} decides how long each bus should dwell beyond its minimum stop time at every station. A decision is triggered at each station arrival for each active bus, producing a high volume of decision epochs throughout the day.

State.. The lower state captures the local context of an individual bus:

$$s_{\mathrm{L}} = \big[\,\text{bus_id},\ \text{station_id},\ \text{hour},\ \text{direction},\ h_{\mathrm{fwd}},\ h_{\mathrm{bwd}},\ \hat{d},\ \Delta h,\ v_1, \ldots, v_K\,\big] \ \in\ \mathbb{R}^{8+K}, \tag{3}$$

where h_{fwd} and h_{bwd} denote the forward and backward headways, $\hat{d}$ is the estimated dwell time, Δh is the deviation of the current headway from h^{target}, and $v_1, \ldots, v_K$ are recent inter-station speeds that encode local traffic conditions.

Action.. The lower action is a scalar holding time:

$$a_{\mathrm{L}} \ \in\ [0,\, 60] \text{ seconds.} \tag{4}$$

Reward and cost.. The immediate lower reward is a function of headway deviation from the target set by the upper level. We define the per-step cost as

$$c(s_{\mathrm{L}}, a_{\mathrm{L}}) = \min\!\Big(\frac{(h_{\mathrm{fwd}} - h^{\mathrm{target}})^2}{(h^{\mathrm{target}})^2},\ 1\Big), \tag{5}$$

which penalizes headway irregularity in a scale-invariant manner and is clipped at 1 for numerical stability.

3.3. Constraint Specification

The bi-level MDP is subject to one hard and one soft constraint, both evaluated through a measurement vector collected at the end of each episode.

Measurement vector.. At the conclusion of a service day, we compute:

$$\mathbf{z} = \big[\,\text{avg_wait_min},\ \text{peak_fleet},\ \text{bunching_rate}\,\big]. \tag{6}$$

Hard constraint (fleet capacity).. The peak number of buses simultaneously on route must not exceed the available fleet:

$$\text{peak_fleet} \ \leq\ N_{\text{fleet}}, \qquad N_{\text{fleet}} = 12. \tag{7}$$

This constraint is enforced structurally: the environment blocks a dispatch if it would violate the fleet limit.

Soft constraint (headway uniformity).. The expected per-step cost must remain below a threshold:

$$\mathbb{E}_{\pi_{\mathrm{U}},\pi_{\mathrm{L}}}\big[\,c(s_{\mathrm{L}}, a_{\mathrm{L}})\,\big] \ \leq\ c_{\text{lim}}, \tag{8}$$

which encourages even spacing of buses along the route. Satisfaction of this constraint is enforced via Lagrangian relaxation within the lower-level policy (see Section 4).

3.4. Timescale Separation

A defining characteristic of the bi-level formulation above is the pronounced separation of timescales between the two decision layers. The upper policy π_{U} acts at dispatch events, which occur roughly every five minutes on average. The lower policy π_{L}, by contrast, acts at every station arrival for every active bus—a frequency on the order of once every 30 seconds system-wide when multiple buses are in service. Consequently, hundreds of lower-level decisions may intervene between two consecutive upper-level decisions, and the two layers never act simultaneously.

This asynchrony distinguishes our setting from the semi-Markov options framework [3] and feudal reinforcement learning [17, 13], which assume that (i) the manager and worker share a common state space or a fixed abstraction thereof, and (ii) the manager selects a subgoal that the worker executes to completion before the manager acts again. Neither assumption holds here: the upper state $s_{\mathrm{U}} \in \mathbb{R}^5$ summarizes fleet-level statistics whereas the lower state $s_{\mathrm{L}} \in \mathbb{R}^{8+K}$ encodes bus-level kinematics, and both policies issue decisions driven by environment events rather than by a synchronous clock or a predefined option duration.

More concretely, a single upper action (a headway triple) remains in effect while many buses traverse multiple stations under independent lower-level control; the upper policy cannot pause and wait for all induced lower decisions to conclude. Standard hierarchical RL methods that propagate value or advantage across levels therefore require modification to handle the fully asynchronous, event-driven interaction between π_{U} and π_{L}. We address this challenge through *Temporal Advantage Propagation*, detailed in Section 4.

4. TransitDuet Framework

This section presents the TransitDuet architecture and its constituent algorithms. Section 4.1 motivates the decoupled dual-network design. Section 4.2 and Section 4.3 detail the upper-level timetable policy and lower-level holding control policy, respectively. Section 4.4 introduces Temporal Advantage Propagation (TAP), the central mechanism that couples the two levels across timescales. Section 4.5 describes the measurement-projection scheme that enforces fleet and service constraints on the upper level. Section 4.6

assembles the full training procedure.

4.1. Architecture Overview

TransitDuet comprises two independent policy networks operating at different timescales and over different state-action spaces . The upper policy π_{U} is a lightweight Gaussian MLP that selects target headways at each dispatch event. The lower policy π_{L} is a DSAC-Lagrangian controller that selects holding durations at every station arrival, with parameters shared across all buses in the fleet.

A key design choice is that the two networks share *no* encoder parameters. This stands in contrast to the shared-body, dual-head architecture of RoboDuet [5], which couples an upper style policy and a lower tracking policy through a common proprioceptive encoder. Shared encoders are appropriate when the two levels observe the same state at each time step—as in synchronous robot locomotion—but are ill-suited here for two reasons. First, s_{U} and s_{L} live in different spaces: s_{U} encodes route-level demand forecasts and fleet status, whereas s_{L} encodes per-bus headway deviation, passenger load, and local occupancy. Second, the two levels are triggered asynchronously. The upper policy acts once per dispatch event (roughly once per h^{target} seconds), while the lower policy acts at every station arrival of every active bus, producing hundreds of decisions per upper-level step. Forcing these heterogeneous signals through a shared trunk would create conflicting gradient directions.

The coupling between the two levels is instead handled entirely through the reward channel, via the Temporal Advantage Propagation mechanism described in section 4.4. This clean separation permits each network to be

trained with the algorithm best suited to its structure: REINFORCE for the episodic, low-dimensional upper policy and soft actor-critic with Lagrangian constraints for the high-frequency, constraint-aware lower policy.

4.2. Upper-Level Timetable Policy

The upper policy π_{U} maps the route-level state s_{U} to a three-dimensional action $a_{\mathrm{U}} = [H_{\mathrm{peak}}, H_{\mathrm{off}}, H_{\mathrm{trans}}]$ specifying target headways for peak, off-peak, and transition periods, respectively. At each dispatch event, the current time-of-day selects the applicable headway component, which is communicated to the lower level as h^{target}.

Network architecture.. The policy is parameterized as a two-layer MLP with 64 hidden units per layer and ReLU activations:

$$\mu_\phi(s_{\mathrm{U}}) = \sigma\big(W_3\,\mathrm{ReLU}(W_2\,\mathrm{ReLU}(W_1 s_{\mathrm{U}} + b_1) + b_2) + b_3\big), \tag{9}$$

where $\sigma(\cdot)$ denotes the element-wise sigmoid function and ϕ collects all weights and biases. The sigmoid output is affinely mapped to the per-dimension action range:

$$a_{\mathrm{U}}^{(k)} = a_{\mathrm{low}}^{(k)} + \mu_\phi^{(k)}(s_{\mathrm{U}}) \cdot \big(a_{\mathrm{high}}^{(k)} - a_{\mathrm{low}}^{(k)}\big), \quad k \in \{1, 2, 3\}. \tag{10}$$

Exploration is achieved by additive Gaussian noise with a learnable log-standard-deviation vector $\log \boldsymbol{\sigma}_\phi$ shared across states:

$$a_{\mathrm{U}} \sim \mathcal{N}\big(\mu_\phi(s_{\mathrm{U}}),\ \mathrm{diag}(\boldsymbol{\sigma}_\phi^2)\big). \tag{11}$$

Policy gradient.. Because the upper level produces only a single action per dispatch event and the episode length is moderate (one simulated day), we

adopt a REINFORCE estimator with normalized returns. Let $\{(s_{\mathrm{U},i}, a_{\mathrm{U},i}, R_i^{\mathrm{U}})\}_{i=1}^{M}$ denote the sequence of M dispatch events in one episode. The policy gradient is

$$\nabla_\phi J(\pi_{\mathrm{U}}) = \frac{1}{M} \sum_{i=1}^{M} \nabla_\phi \log \pi_{\mathrm{U}}(a_{\mathrm{U},i} \mid s_{\mathrm{U},i}) \cdot \hat{A}_{\mathrm{U},i}, \tag{12}$$

where $\hat{A}_{\mathrm{U},i}$ is the TAP-augmented advantage defined in section 4.4 and returns are normalized to zero mean and unit variance within the episode to reduce gradient variance.

4.3. Lower-Level Holding Control Policy

The lower policy π_{L} selects a holding duration $a_{\mathrm{L}} \in [0, a_{\max}]$ at each station arrival, where $a_{\max} = 60\,\mathrm{s}$. It is trained with a distributional soft actor-critic augmented by a Lagrangian cost term for headway-deviation constraint enforcement [7]. All buses share the same policy parameters, enabling fleet-wide generalization.

Gaussian policy with tanh squashing.. A two-layer MLP with 32 hidden units per layer and ReLU activations produces a mean $\mu_\psi(s_{\mathrm{L}})$ and log-standard-deviation $\log \sigma_\psi(s_{\mathrm{L}})$. The raw sample $u \sim \mathcal{N}(\mu_\psi, \sigma_\psi^2)$ is squashed through a tanh transform and rescaled:

$$a_{\mathrm{L}} = \frac{a_{\max}}{2}\big(\tanh(u) + 1\big). \tag{13}$$

The log-probability accounts for the change of variables:

$$\log \pi_{\mathrm{L}}(a_{\mathrm{L}} \mid s_{\mathrm{L}}) = \log \mathcal{N}(u;\, \mu_\psi,\, \sigma_\psi^2) - \log\big(1 - \tanh^2(u)\big) - \log \frac{a_{\max}}{2}. \tag{14}$$

Twin Q-networks.. Two independent critics $Q_{\omega_1}(s_{\mathrm{L}}, a_{\mathrm{L}})$ and $Q_{\omega_2}(s_{\mathrm{L}}, a_{\mathrm{L}})$, each a two-layer MLP with 32 hidden units, are trained to minimize the clipped double-Q Bellman error:

$$\mathcal{L}_Q(\omega_j) = \mathbb{E}_{(s_{\mathrm{L}}, a_{\mathrm{L}}, r, s'_{\mathrm{L}}) \sim \mathcal{D}} \Big[\big(Q_{\omega_j}(s_{\mathrm{L}}, a_{\mathrm{L}}) - y \big)^2 \Big], \quad j \in \{1, 2\}, \tag{15}$$

where the target is

$$y = r + \gamma \Big(\min_{j=1,2} Q_{\bar{\omega}_j}(s'_{\mathrm{L}}, a'_{\mathrm{L}}) - \alpha \log \pi_{\mathrm{L}}(a'_{\mathrm{L}} \mid s'_{\mathrm{L}}) \Big), \quad a'_{\mathrm{L}} \sim \pi_{\mathrm{L}}(\cdot \mid s'_{\mathrm{L}}), \tag{16}$$

and $\bar{\omega}_j$ denotes target-network parameters updated via soft interpolation with coefficient $\tau = 0.005$.

Cost Q-network and Lagrangian relaxation.. A separate cost critic $Q^c_\xi(s_{\mathrm{L}}, a_{\mathrm{L}})$ estimates the expected cumulative headway-deviation cost. The lower-level policy objective balances reward maximization, entropy regularization, and constraint satisfaction:

$$\mathcal{L}_{\pi_{\mathrm{L}}}(\psi) = \mathbb{E}_{s_{\mathrm{L}} \sim \mathcal{D},\, a_{\mathrm{L}} \sim \pi_{\mathrm{L}}} \Big[\alpha \log \pi_{\mathrm{L}}(a_{\mathrm{L}} \mid s_{\mathrm{L}}) - \min_j Q_{\omega_j}(s_{\mathrm{L}}, a_{\mathrm{L}}) + \lambda Q^c_\xi(s_{\mathrm{L}}, a_{\mathrm{L}}) \Big], \tag{17}$$

where $\lambda \geq 0$ is the Lagrange multiplier. The dual variable is updated to satisfy the cost-limit constraint c_{lim}:

$$\lambda \leftarrow \exp\Big[\mathrm{clamp}\big(\log \lambda + \eta_\lambda (c_{\mathrm{lim}} - \mathbb{E}[Q^c_\xi]),\ -10,\ 5 \big) \Big]. \tag{18}$$

Updating in log-space ensures $\lambda > 0$ and the clamp prevents numerical instability.

Auto-entropy tuning.. The temperature α is adjusted to maintain a target entropy $\bar{\mathcal{H}} = -\dim(a_{\mathrm{L}})$:

$$\alpha \leftarrow \min\Big(\alpha_{\max},\ \exp\big(\log \alpha - \eta_\alpha \left(\log \pi_{\mathrm{L}}(a_{\mathrm{L}} \mid s_{\mathrm{L}}) + \bar{\mathcal{H}} \right) \big) \Big), \tag{19}$$

with $\alpha_{\max} = 0.3$ to prevent entropy collapse from dominating the Q-values in the early training phase.

4.4. Temporal Advantage Propagation

Temporal Advantage Propagation (TAP) is the central mechanism by which TransitDuet couples the upper and lower levels despite their operating on different timescales with different state-action spaces. We first review the synchronous cross-advantage of **(author?)** [5], then present the asynchronous generalization that constitutes our primary contribution.

4.4.1. Synchronous Cross-Advantage (Background)

In RoboDuet, the upper and lower policies share a common state clock. At each time step t, both policies execute, and both advantages $A_{\mathrm{U},t}$ and $A_{\mathrm{L},t}$ are well-defined. The augmented objective for the upper policy is

$$\mathcal{L}^{\mathrm{U}}_{\mathrm{RoboDuet}} = -\big(A_{\mathrm{U},t} + \beta\, A_{\mathrm{L},t}\big) \cdot \rho_t, \tag{20}$$

where ρ_t is the importance-sampling ratio from PPO. This formulation requires that $A_{\mathrm{U},t}$ and $A_{\mathrm{L},t}$ are computed at the same time step t, which is possible only when both levels share a common clock and state representation.

4.4.2. Asynchronous Generalization

In bus operations, the upper policy acts at dispatch event i (when bus i departs the terminal), while the lower policy acts at every station arrival of every bus throughout the simulation. Let $\mathrm{trip}(i)$ denote the set of lower-level transitions that occur between dispatch event i and dispatch event $i{+}1$—that

is, all station arrivals across the fleet during the lifecycle of the trip initiated by dispatch i.

Forward direction: lower → upper.. We augment the upper advantage at dispatch i with the mean lower-level reward accumulated during trip(i):

$$\tilde{A}_{\mathrm{U}}[i] = A_{\mathrm{U}}[i] + \beta \cdot \overline{R}_{\mathrm{L}}^{(i)}, \quad \text{where} \quad \overline{R}_{\mathrm{L}}^{(i)} = \frac{1}{|\mathrm{trip}(i)|} \sum_{j \in \mathrm{trip}(i)} r_{\mathrm{L},j}. \tag{21}$$

The mean operator, rather than a sum, normalizes for the variable number of lower-level transitions per trip, which fluctuates with the dispatched headway: a short headway produces fewer intermediate station arrivals than a long one.

Reverse direction: upper → lower.. The lower-level training reward for each transition $j \in \mathrm{trip}(i)$ is augmented with the upper-level reward of the initiating dispatch:

$$\tilde{r}_{\mathrm{L},j} = r_{\mathrm{L},j} + \beta \cdot R_{\mathrm{U}}^{(i)}, \quad \forall\, j \in \mathrm{trip}(i). \tag{22}$$

This injects global timetable-quality feedback into the local holding decisions, enabling the lower policy to distinguish states that arise from good timetables (worth exploiting) from those arising from poor ones (requiring compensatory holding).

Physical intuition.. Suppose the upper policy dispatches a bus with an unrealistically long headway $h^{\mathrm{target}} = 180\,\mathrm{s}$ during peak demand. The lower policy will encounter overcrowded stops and large headway deviations, generating consistently negative $r_{\mathrm{L},j}$. Through the forward direction of TAP, the mean of these negative rewards feeds back into $\tilde{A}_{\mathrm{U}}$, penalizing the dispatch

decision and guiding the upper policy toward more feasible headways. Conversely, if the lower policy consistently fails to hold buses despite receiving a reasonable headway target, the reverse direction of TAP injects the positive upper reward into lower transitions, signaling that the timetable context was favorable and encouraging the lower policy to improve its holding behavior.

Relationship to synchronous cross-advantage.. When both levels share a common clock—i.e., $|\mathrm{trip}(i)| = 1$ for all i and $r_{\mathrm{L},j} = A_{\mathrm{L},t}$—eq. (21) reduces to $\tilde{A}_{\mathrm{U}} = A_{\mathrm{U}} + \beta\, A_{\mathrm{L}}$, recovering eq. (20). TAP is thus a strict generalization of the synchronous cross-advantage to multi-timescale, asynchronous bi-level systems.

4.4.3. Coupling Coefficient Schedule

Coupling the two levels from the start of training is harmful: the lower policy has not yet learned reasonable holding behavior, so its rewards carry no useful signal for the upper policy, and vice versa. We therefore anneal β through three stages:

$$\beta(e) = \begin{cases} 0, & e \le e_1, \\ \beta_{\max} \cdot \dfrac{e - e_1}{e_2 - e_1}, & e_1 < e < e_2, \\ \beta_{\max}, & e \ge e_2, \end{cases} \tag{23}$$

where e is the episode index, $e_1 = 50$, $e_2 = 150$, and $\beta_{\max} = 0.5$. Stage 1 ($e \le e_1$) is a lower-only warmup in which the upper policy is frozen and π_{L} trains against a fixed default timetable. Stage 2 ($e_1 < e < e_2$) activates the upper policy and linearly ramps β from zero to $\beta_{\max}$, progressively strengthening the cross-level coupling. Stage 3 ($e \ge e_2$) maintains full coupling at $\beta = \beta_{\max}$.

Algorithm 1 Temporal Advantage Propagation (one episode)

Require: Upper trajectory $\{(s_{\mathrm{U},i}, a_{\mathrm{U},i}, R_{\mathrm{U},i})\}_{i=1}^{M}$, lower transitions $\{(s_{\mathrm{L},j}, a_{\mathrm{L},j}, r_{\mathrm{L},j})\}_{j=1}^{N}$, trip mapping $\mathrm{trip}(\cdot)$, coupling coefficient β

Ensure: Augmented upper advantages $\{\tilde{A}_{\mathrm{U}}[i]\}$, augmented lower rewards $\{\tilde{r}_{\mathrm{L},j}\}$

// **Forward: lower → upper**
for $i = 1, \ldots, M$ **do**
  $\overline{R}_{\mathrm{L}}^{(i)} \leftarrow \frac{1}{|\mathrm{trip}(i)|} \sum_{j \in \mathrm{trip}(i)} r_{\mathrm{L},j}$
  Compute baseline: $\hat{A}_{\mathrm{U},i} \leftarrow R_{\mathrm{U},i} - \bar{R}_{\mathrm{U}}$ {normalized return}
  $\tilde{A}_{\mathrm{U}}[i] \leftarrow \hat{A}_{\mathrm{U},i} + \beta \cdot \overline{R}_{\mathrm{L}}^{(i)}$
end for

// **Reverse: upper → lower**
for $j = 1, \ldots, N$ **do**
  Let $i(j)$ be the dispatch index such that $j \in \mathrm{trip}(i(j))$
  $\tilde{r}_{\mathrm{L},j} \leftarrow r_{\mathrm{L},j} + \beta \cdot R_{\mathrm{U},i(j)}$
end for

The linear ramp is intentionally simple; we found empirically that more elaborate schedules (cosine annealing, step functions) produced no consistent improvement while introducing additional hyperparameters.

4.4.4. Algorithm

The TAP computation for one episode is summarized in algorithm 1.

4.5. Measurement Projection for Fleet Constraint

The upper policy must satisfy operational constraints on fleet size, average passenger waiting time, and bus bunching rate. Rather than adding penalty terms to the reward—which requires manual tuning and conflates constraint satisfaction with reward maximization—we adopt a measurement-projection approach inspired by the ApproPO framework [8], simplified to avoid the nested best-response oracle that ApproPO requires.

Measurement vector.. At the end of each episode, we compute a three-dimensional measurement vector:

$$\mathbf{z} = \left[\bar{w},\ n_{\text{peak}},\ b_{\text{rate}}\right] \in \mathbb{R}^3, \tag{24}$$

where $\bar{w}$ is the average passenger waiting time, n_{peak} is the peak fleet size (maximum number of buses simultaneously in service), and b_{rate} is the bunching rate (fraction of arrivals with headway below a threshold).

Weight vector and reward modulation.. A weight vector $\boldsymbol{\theta} \in \mathbb{R}^3$, initialized as $\boldsymbol{\theta}_0 = [-0.5,\ -0.3,\ -0.2]$, modulates the upper-level reward:

$$R_{\text{U}} = \boldsymbol{\theta}^\top \mathbf{p}, \quad \text{where} \quad \mathbf{p} = \left[\bar{w},\ (\max(0,\ n_{\text{peak}} - N_{\text{fleet}}))^2,\ b_{\text{rate}}\right]. \tag{25}$$

The quadratic penalty on fleet overshoot $(\max(0,\ n_{\text{peak}} - N_{\text{fleet}}))^2$ ensures that small violations incur proportionally less cost than large ones, providing a smooth gradient.

Online gradient descent update.. The weight vector is updated by online gradient descent (OGD) on a loss that drives each measurement toward its target:

$$\ell_t = \left[-\bar{w}_t,\ -\max(0,\ n_{\text{peak},t} - N_{\text{fleet}}),\ -b_{\text{rate},t}\right], \tag{26}$$

$$\boldsymbol{\theta}_{t+1} = \Pi_{\mathcal{B}}\big(\boldsymbol{\theta}_t + \eta_t\, \ell_t\big), \quad \eta_t = \frac{\eta_0}{\sqrt{t}}, \tag{27}$$

where $\Pi_{\mathcal{B}}$ projects onto the unit ball $\{\boldsymbol{\theta} : \|\boldsymbol{\theta}\|_2 \leq 1\}$ and η_0 is the base learning rate. The $1/\sqrt{t}$ decay guarantees that the cumulative constraint violation grows sublinearly, yielding an $O(\sqrt{T})$ regret bound standard in online convex optimization.

The simplification relative to ApproPO is deliberate: because the upper policy is trained via REINFORCE with the $\boldsymbol{\theta}$-modulated reward, the policy update itself serves as the approximate best-response oracle, eliminating the inner optimization loop that ApproPO requires.

4.6. Training Procedure

Algorithm 2 presents the full TransitDuet training loop, which coordinates the three-stage β schedule with the upper and lower policy updates, TAP, and measurement projection.

Stage 1: lower-only warmup ($e \leq e_1 = 50$).. The upper policy is frozen and all buses are dispatched with a fixed default headway of $h^{\text{target}} = 360\,\text{s}$. The lower policy trains in isolation via DSAC-Lagrangian, building a competent holding controller before any cross-level coupling is introduced. This warmup is critical: without it, the upper policy receives TAP feedback from a random lower policy, which amounts to noise and destabilizes upper-level learning.

Stage 2: ramp-up ($e_1 < e < e_2$).. The upper policy is activated and begins selecting headways. The coupling coefficient β increases linearly from 0 to $\beta_{\max} = 0.5$, progressively strengthening the bidirectional reward augmentation. Both policies are updated at their respective frequencies: the lower

policy every $f_{\mathrm{L}} = 5$ station arrivals with batch size $B = 2048$, and the upper policy once per episode. The measurement-projection weight $\boldsymbol{\theta}$ is also updated at each episode boundary via OGD.

Stage 3: full coupling ($e \geq e_2 = 150$).. The coupling coefficient is held at $\beta_{\max} = 0.5$ for the remainder of training. Both policies continue to update, and the OGD weight $\boldsymbol{\theta}$ continues to adapt, gradually shifting the upper reward emphasis toward whichever constraint is most frequently violated.

Computational cost.. The lower policy update dominates computation due to the twin Q-network and cost Q-network gradient steps. Each lower update involves five backward passes (two Q-critics, one cost critic, one policy, one entropy temperature), performed on GPU-batched samples from $\mathcal{D}$. The upper policy update is negligible by comparison: a single REINFORCE gradient step over $M \approx 30$–50 dispatch events. The TAP computation adds only $O(M + N)$ scalar operations per episode and is not a bottleneck.

Convergence considerations.. The three-stage schedule ensures that coupling is introduced only after both policies have reached a reasonable performance baseline. The linear β ramp avoids the discontinuity that a step-function activation would introduce, which in preliminary experiments caused oscillatory instabilities in both the upper and lower value estimates. We did not observe sensitivity to the precise values of e_1 and e_2 within a factor of two of the defaults, but setting e_1 too small (before the lower policy converges to a stable holding strategy) consistently degraded final performance, as confirmed by the ablation study in section 5.

5. Experiments

We evaluate TransitDuet on an event-driven bus microsimulation and compare against five ablated or simplified baselines. All numerical results are pending and marked as **[TBD]**.

5.1. Experimental Setup

Simulation environment.. We implement an event-driven bus simulation modeling a single bidirectional route with 22 stations (11 per direction), serving 264 scheduled trips per day. Passenger demand follows origin–destination (OD) matrices with time-varying arrival rates calibrated to reproduce morning and evening peaks. The simulation advances at a resolution of $\Delta t = 1\,\mathrm{s}$. Passenger arrivals and alighting events are updated every $20\,\mathrm{s}$, while the upper-level route planner is queried every $300\,\mathrm{s}$ ($5\,\mathrm{min}$) to issue dispatch decisions. Travel times between consecutive stations are drawn from log-normal distributions whose parameters vary by time of day and direction, with a route stochasticity parameter σ controlling variance (default $\sigma = 1.5$).

Methods.. We compare six variants, labeled **A**–**F**:

A. TransitDuet (full): the complete framework with TAP, $\boldsymbol{\theta}$-OGD, Lagrangian penalties, and cosine-annealed β schedule.

B. No TAP ($\beta = 0$ always): both levels train independently; the cross-advantage augmentation is disabled.

C. No $\boldsymbol{\theta}$-OGD (fixed reward weights): the measurement-projection constraint on fleet utilization is replaced by fixed, hand-tuned reward weights.

D. No Lagrangian ($c_{\text{lim}} = \infty$): the headway-regularity constraint at the lower level is removed; the agent relies on reward shaping alone.

E. Fixed Timetable (π_{U} disabled): the upper policy is replaced by a static timetable with $h^{\text{target}} = 360\,\text{s}$ throughout the day; only the lower-level holding controller is trained.

F. Flat RL: a single-level distributional soft actor-critic (DSAC) agent that jointly selects dispatch and holding actions without hierarchical decomposition.

Metrics.. We report four primary metrics averaged over the final 50 evaluation episodes:

- *Average passenger wait time* (min): the mean time between a passenger's arrival at a station and boarding.
- *Bunching rate* (%): the fraction of station arrivals where the forward headway is less than half the target headway.
- *Peak fleet size*: the maximum number of buses simultaneously in service during an episode.
- *Episode reward*: the undiscounted cumulative reward over one full day of operation.

Training protocol.. Each method is trained for 300 episodes with three independent random seeds. We report means and standard deviations across seeds.

Table 1: Hyperparameters for TransitDuet.

Component	**Parameter**	**Value**
Upper policy	Learning rate	1×10^{-4}
Lower policy	Learning rate	1×10^{-5}
Lagrangian	Dual variable learning rate	1×10^{-3}
Training	Batch size	2048
	Discount factor γ	0.99
	Soft-update coefficient τ	0.005
TAP schedule	$\beta_{\max}$	0.5
	Warmup episodes	50
	Ramp episodes	100
Fleet	N_{fleet}	12
Constraint	c_{lim}	0.15

Hyperparameters.. Table 1 summarizes the key hyperparameters. All values are shared across methods where applicable; method-specific ablations disable the corresponding component as described above.

5.2. Main Results

Table 2 presents the performance of all six methods.

TransitDuet vs. Fixed Timetable (A vs. E).. Replacing the static 360 s headway with a learned, time-varying target reduces average wait time by **[TBD]**% and bunching rate by **[TBD]**%. This confirms that adaptive timetable planning provides headroom that holding control alone cannot exploit.

Table 2: Performance comparison across six methods (mean ± std over 3 seeds). Best result in **bold**.

Method	**Wait (min)**	**Bunching (%)**	**Peak Fleet**	**Reward**
A. TransitDuet	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
B. No TAP	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
C. No $\boldsymbol{\theta}$-OGD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
D. No Lagrangian	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
E. Fixed Timetable	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
F. Flat RL	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TransitDuet vs. Flat RL (A vs. F).. The hierarchical decomposition outperforms the monolithic agent by **[TBD]**% on wait time. We attribute this to the difficulty of learning dispatch and holding jointly in a single action space with heterogeneous timescales, which manifests as high variance in training and suboptimal fleet utilization.

Effect of inter-level coupling (A vs. B).. Disabling TAP ($\beta = 0$) degrades wait time by **[TBD]**%, demonstrating that the cross-advantage signal is essential for coordinating the two levels rather than merely training them in parallel.

Effect of constraint mechanisms (A vs. C and A vs. D).. Removing $\boldsymbol{\theta}$-OGD (Method C) leads to a peak fleet size of **[TBD]**, exceeding the budget $N_{\text{fleet}} = 12$ in **[TBD]**% of episodes. Removing the Lagrangian penalty (Method D) inflates the bunching rate to **[TBD]**%, indicating that reward shaping alone is insufficient to maintain headway regularity.

[TBD: training curves A vs. B, x-axis=episode, y-axis=avg wait time]

Figure 1: Training curves comparing TransitDuet (A) and No TAP (B). The shaded region indicates ± 1 standard deviation across 3 seeds. The vertical dashed line marks the onset of β ramping at episode 50.

[TBD: peak fleet size A vs. C, x-axis=episode, y-axis=peak fleet]

Figure 2: Peak fleet size during training for TransitDuet (A) and No $\boldsymbol{\theta}$-OGD (C). The horizontal dashed line marks the fleet budget $N_{\text{fleet}} = 12$.

5.3. Ablation Study

We isolate the contribution of each component through pairwise ablations.

TAP coupling (A vs. B).. Figure 1 plots the training curves of average wait time for Methods A and B. TransitDuet converges to **[TBD]** min by episode **[TBD]**, whereas the no-TAP variant plateaus at **[TBD]** min. The gap widens precisely when β begins ramping (episode 50), suggesting that TAP accelerates credit assignment across levels.

$\boldsymbol{\theta}$-OGD projection (A vs. C).. Figure 2 shows peak fleet size over training episodes. Without $\boldsymbol{\theta}$-OGD, the upper policy over-dispatches during the peak, reaching **[TBD]** buses simultaneously. The projection mechanism keeps peak fleet at or below $N_{\text{fleet}} = 12$ after episode **[TBD]**.

Lagrangian penalty (A vs. D).. Figure 3 presents the distribution of headway deviations $|h - h^{\text{target}}|/h^{\text{target}}$ at convergence. Method D exhibits a heavy right tail with **[TBD]**% of observations exceeding a 50% relative deviation, compared to **[TBD]**% for TransitDuet.

[TBD: headway deviation histogram A vs. D]

Figure 3: Distribution of relative headway deviations at convergence for TransitDuet (A) and No Lagrangian (D). Lower and more concentrated is better.

[TBD: 3-line plot of θ_{wait}, θ_{headway}, θ_{fleet} over episodes]

Figure 4: Evolution of $\boldsymbol{\theta}$-OGD weight components during training. The fleet weight θ_{fleet} peaks early and decays as the constraint is internalized.

5.4. Mechanism Analysis

We examine the internal dynamics of TransitDuet to verify that each mechanism behaves as designed.

$\boldsymbol{\theta}$ weight evolution.. Figure 4 plots the three components of the OGD weight vector $\boldsymbol{\theta} = [\theta_{\text{wait}}, \theta_{\text{headway}}, \theta_{\text{fleet}}]$ over training. Early in training, θ_{fleet} dominates as the policy learns to satisfy the fleet constraint; after approximately episode **[TBD]**, θ_{wait} and θ_{headway} increase as the policy shifts toward service quality optimization within the feasible region.

Lagrangian multiplier λ convergence.. Figure 5 shows the Lagrangian dual variable λ for the headway-regularity constraint. Starting from $\lambda_0 = 0$, the multiplier rises sharply during early training when headway deviations are large, then converges to a stable value of **[TBD]** as the lower policy learns to regulate headways within the tolerance $c_{\text{lim}} = 0.15$.

Learned target headways.. Figure 6 visualizes the upper policy's output—H_{peak}, H_{off}, and H_{trans}—over training episodes, each averaged across dispatch events in the corresponding demand period. The policy learns to compress peak headways to approximately **[TBD]** s (high frequency) while relaxing off-

[TBD: λ vs. episode]

Figure 5: Convergence of the Lagrangian multiplier λ during training. Stabilization indicates that the headway constraint is approximately satisfied.

[TBD: 3-line plot of H_{peak}, H_{off}, H_{trans} over episodes]

Figure 6: Learned target headways over training. The upper policy differentiates peak, off-peak, and transition periods, converging to demand-responsive dispatch intervals.

peak headways to **[TBD]** s, a pattern consistent with operational practice. Transition headways settle to intermediate values of **[TBD]** s, smoothing the shift between demand regimes.

β schedule effect.. To confirm the necessity of the warmup-then-ramp β schedule, we train a variant with $\beta = \beta_{\max}$ from episode 1. This aggressive coupling causes training instability, with wait time oscillating between **[TBD]** and **[TBD]** min over the final 50 episodes, compared to a stable **[TBD]** $\pm$ **[TBD]** min for the annealed schedule. The warmup period allows each policy to establish a reasonable baseline before cross-advantage signals introduce inter-level gradients.

5.5. Generalization Across Demand Stochasticity

We test whether a policy trained at route stochasticity $\sigma = 1.5$ generalizes to unseen demand conditions by evaluating on $\sigma \in \{0.5, 1.0, 2.0, 3.0\}$ without retraining. Table 3 reports the results.

Performance degrades gracefully as σ increases beyond the training distribution. At $\sigma = 2.0$, wait time rises by **[TBD]**% relative to $\sigma = 1.5$, while bunching rate increases by **[TBD]** percentage points. Even at $\sigma =$

Table 3: Cross-stochasticity generalization. The model is trained at $\sigma = 1.5$ (shaded) and evaluated at other σ values without fine-tuning.

σ	**Wait (min)**	**Bunching (%)**	**Peak Fleet**	**Reward**
0.5	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
1.0	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
1.5	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
2.0	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
3.0	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

3.0—double the training stochasticity—TransitDuet maintains a wait time of **[TBD]** min, which remains **[TBD]**% lower than the Fixed Timetable baseline evaluated at the same σ. Conversely, at $\sigma = 0.5$ the policy achieves its best performance (**[TBD]** min wait), confirming that the learned strategy is conservative enough to handle reduced variance without over-dispatching. The fleet constraint ($N_{\text{fleet}} = 12$) is satisfied across all tested σ values, validating the robustness of the $\boldsymbol{\theta}$-OGD projection mechanism.

6. Discussion

Why TAP works.. Temporal Advantage Propagation succeeds because it closes a feedback loop that is absent when the two levels train independently. Without TAP, the upper policy receives reward signals only from its own sparse dispatch events, learning that a particular target headway yields a certain fleet cost—but remaining oblivious to whether the lower policy can actually realize that headway under stochastic conditions. Conversely, the lower policy optimizes holding actions against a headway target it treats

as exogenous, unable to signal the upper level that a target is infeasible or wasteful. TAP bridges this gap by propagating discounted lower-level rewards back to the dispatch decision that spawned them, and by augmenting the lower advantage with the upper-level value of the dispatch context. The cosine-annealed β schedule ensures that these cross-level signals are introduced only after each policy has developed a reasonable standalone strategy, preventing the gradient noise of one immature policy from destabilizing the other.

Connection to RoboDuet.. TransitDuet draws direct inspiration from the RoboDuet framework of **(author?)** [5], which introduced cross-advantage augmentation for bi-level robot locomotion control. Table 4 makes the structural analogy explicit. The key generalization is temporal: RoboDuet's two levels share a common state clock, permitting direct advantage summation at each time step, whereas TransitDuet operates across asynchronous timescales that require trip-lifecycle aggregation.

Limitations.. Several limitations qualify the present results and motivate future work.

Single-route scope. TransitDuet is evaluated on a single bidirectional corridor. Real transit networks involve multiple interacting routes with shared terminals, transfer passengers, and fleet rebalancing across lines. Extending the framework to a multi-route setting would require the upper policy to allocate a shared fleet across routes, substantially expanding the action space and introducing inter-route coordination constraints.

Simplified demand model. Passenger arrivals follow stationary OD rates within each time period. In practice, demand is non-stationary, exhibits day-

Table 4: Structural analogy between RoboDuet and TransitDuet.

Aspect	**RoboDuet**	**TransitDuet**
Upper policy	Locomotion style selector	Timetable planner (π_{U})
Lower policy	Joint-level tracking controller	Holding controller (π_{L})
Upper action	Style parameters	Target headways $[H_{\mathrm{peak}}, H_{\mathrm{off}}, H_{\mathrm{trans}}]$
Lower action	Joint torques	Holding duration (s)
Coupling	Synchronous cross-advantage	Asynchronous TAP
Timescale	Shared clock (same Δt)	Heterogeneous (300 s vs. 1 s)
Constraint	Joint-limit penalties	Fleet ($\boldsymbol{\theta}$-OGD) + headway (Lagrangian)
β schedule	Fixed β	Cosine-annealed warmup

of-week and seasonal variation, and responds endogenously to service quality (e.g., passengers abandoning a stop after excessive waits). Incorporating demand elasticity and real-time demand forecasting would improve realism but introduce additional model complexity.

No real-world deployment. All experiments are conducted in a stochastic microsimulation. While the simulator reproduces key phenomena—bunching, load-dependent dwell times, stochastic travel times—it abstracts away traffic signal interactions, driver behavior heterogeneity, and vehicle breakdowns. Sim-to-real transfer would require domain randomization or a calibrated digital twin.

Computational cost. Each episode requires approximately 7 s of environment simulation time plus training overhead. At 300 episodes with 3 seeds, the total wall-clock time is approximately **[TBD]** hours on a single GPU.

While acceptable for research, real-time retraining for schedule disruptions would require further acceleration, for instance through model distillation or warm-starting from a pre-trained policy.

Scalability considerations.. Scaling to multi-route networks introduces two challenges. First, the upper-level action space grows combinatorially with the number of routes, since headway targets must be set jointly to respect shared fleet constraints. Factored action representations or attention-based architectures over route embeddings could mitigate this. Second, the lower-level parameter sharing, which currently treats all buses on a single route identically, would need to incorporate route-specific context to generalize across heterogeneous lines. Graph neural networks operating on the transit network topology offer a promising direction for encoding such structure.

7. Conclusion

This paper introduced TransitDuet, a bi-level reinforcement learning framework that unifies bus timetable planning and real-time holding control within a single trainable system. The upper policy learns demand-responsive target headways at dispatch events, while the lower policy regulates individual holding actions at station arrivals—two tasks that have traditionally been addressed in isolation despite their tight operational coupling.

The central technical contribution is Temporal Advantage Propagation (TAP), an asynchronous cross-advantage mechanism that aggregates lower-level feedback over variable-length trip lifecycles and propagates it to the upper-level dispatch decisions that initiated them. TAP generalizes the synchronous cross-advantage of RoboDuet to settings where the two control

levels operate on fundamentally different timescales and observe distinct state-action spaces. A cosine-annealed coupling schedule prevents premature inter-level interference during early training. Operational constraints are handled through complementary mechanisms: online gradient descent projection ($\boldsymbol{\theta}$-OGD) enforces fleet-size feasibility at the upper level, while Lagrangian relaxation maintains headway regularity at the lower level.

Experiments on a stochastic bus corridor microsimulation with 22 stations and 264 daily trips show that TransitDuet reduces mean passenger waiting time by **[TBD]**% and bunching rate by **[TBD]**% relative to the strongest single-level baseline, while satisfying the fleet constraint in **[TBD]**% of evaluation episodes. Ablation studies confirm that each component—TAP, $\boldsymbol{\theta}$-OGD, and Lagrangian penalties—contributes to this result; removing any one of them leads to measurable degradation in at least one metric. The learned policy generalizes across unseen levels of demand stochasticity without retraining, maintaining a **[TBD]**% wait-time advantage even at double the training variance.

Several directions merit future investigation. First, extending TransitDuet to multi-route networks with shared fleets and transfer coordination would test whether the hierarchical structure scales to the combinatorial action spaces of realistic transit systems. Second, real-world deployment—potentially through a calibrated digital twin with domain randomization—would validate the framework's robustness to phenomena not captured by the current simulator, including traffic signal interactions and driver behavior heterogeneity. Third, integrating real-time demand forecasting and modeling passenger demand elasticity would close the loop between service qual-

ity and ridership. Finally, coupling TransitDuet with passenger information systems—broadcasting predicted arrival times based on the learned holding policy—could amplify the welfare benefits by reducing perceived waiting time alongside actual waiting time.

CRediT authorship contribution statement

Anonymous Authors: Conceptualization, Methodology, Software, Validation, Writing – Original Draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Code and data will be made available upon publication.

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Appendix A. Hyperparameter Details

Table 1 lists all hyperparameters used in the experiments. Values correspond to the default configuration; no per-experiment tuning was performed.

Appendix B. Environment Details

The simulation models a single bidirectional bus route calibrated from operational data of an urban bus corridor.

Route topology.. The route comprises M stations connected by inter-station segments. Each direction (upstream and downstream) traverses all M stations in sequence, with terminal turnaround at each end. Station positions and inter-station distances are read from operational records (`stop_news.xlsx`), and segment-level speed profiles are specified per hour of day (`route_news.xlsx`).

Stochastic travel speed.. Inter-station travel times are driven by a stochastic speed model. At each route-state update, the speed limit on segment $(i, i{+}1)$ is drawn as

$$v_i = \min\bigl(V_{\max},\ \max\bigl(0,\ \lfloor \ln \mathcal{LN}(\mu_i(t),\, \sigma) \rfloor\bigr)\bigr), \tag{B.1}$$

where $\mu_i(t)$ is the historical mean log-speed for segment i at hour t, $\sigma = 1.5$ is the log-normal dispersion parameter (`route_sigma`), and $V_{\max}$ is the segment maximum speed. This produces realistic speed variability that induces headway deviations even under a well-designed timetable.

Passenger generation.. Passenger arrivals follow a Poisson process governed by time-varying origin–destination (OD) matrices (`passenger_OD.xlsx`). The OD data specifies hourly demand rates between all station pairs. At each simulation second, for every origin station with non-zero demand to a given destination, a Poisson draw with rate $\lambda_{od}/3600$ determines the number of new passengers. Each passenger records an appearance time, an origin, and a destination; waiting time is measured from appearance to boarding.

Bus operations.. Each bus maintains a state machine with four phases:

1. **Holding** — the bus is held at a station for the duration specified by the lower-level policy (0–60 s).

2. **Dwelling** — passengers alight and board. Boarding proceeds until the bus reaches capacity ($C = 50$ passengers) or no more waiting passengers have a valid destination downstream.

3. **Travel** — the bus moves toward the next station at the current stochastic speed. Acceleration (3 m/s^2) and deceleration (5 m/s^2) are modeled at departure and arrival.

4. **Waiting for action** — upon arriving at a station, the bus waits for the next holding action from the lower-level policy.

Forward and backward headways are computed with respect to the preceding and following buses in the same direction, and the headway deviation from the target set by the upper policy defines the per-step cost (Equation (5)).

Timetable structure.. A service day consists of 264 trips alternating between the two directions, truncated from a full operational schedule to keep simulation tractable. At most 25 buses may be concurrently active (`max_agent_num`). The upper policy controls the dispatch headway—the time gap between consecutive launches in the same direction—which also serves as the target headway h^{target} communicated to the lower-level controller.

Episode termination.. An episode ends when all 264 trips have been launched and every bus has completed its trip (i.e., no bus remains on-route). At this point the measurement vector $\mathbf{z}$ (Equation (6)) is computed for the ApproPO $\boldsymbol{\theta}$-OGD update.

Appendix C. Network Architectures

All networks use ReLU activations and are trained with Adam. Weight initialization follows PyTorch defaults unless otherwise noted.

Upper policy π_{U} (Gaussian).. A two-layer MLP maps the 5-dimensional upper state to headway parameters:

Layer	**Shape**	**Activation**
Linear	$5 \to 64$	ReLU
Linear	$64 \to 64$	ReLU
Mean head	$64 \to 3$	Sigmoid $\to$ affine rescale to [`action_low`, `action_high`]
Log-std head	$64 \to 3$	Clamp$[-5, 0]$

The mean and log-std heads are initialized with zero bias to start near the midpoint of the action range. During rollout, actions are sampled from

$\mathcal{N}(\mu, \sigma^2)$ and clamped to the valid range. For policy-gradient evaluation, the reparameterization trick is used and log-probabilities are summed across the 3 action dimensions.

Lower policy π_{L} (Squashed Gaussian).. A two-layer MLP with tanh squashing maps the lower state to a scalar holding time:

Layer	**Shape**	**Activation**
Linear	$(8{+}K) \to 32$	ReLU
Linear	$32 \to 32$	ReLU
Mean head	$32 \to 1$	—
Log-std head	$32 \to 1$	Clamp$[-20, 2]$

The raw sample $z \sim \mathcal{N}(\mu, \sigma^2)$ is squashed via $a = \tanh(z) \times 60$ to produce a holding time in $[0, 60]$ s. Log-probabilities include the standard tanh Jacobian correction [7]. Weights in the output heads are initialized uniformly in $[-3{\times}10^{-3}, 3{\times}10^{-3}]$. The same network is shared across all buses (parameter sharing), with bus-specific behavior arising from heterogeneous input states.

Twin reward Q-network Q_1, Q_2.. Two independent MLPs implement clipped double Q-learning for the lower level:

Layer	**Shape**	**Activation**
Linear	$(8{+}K{+}1) \to 32$	ReLU
Linear	$32 \to 32$	ReLU
Linear	$32 \to 1$	—

Each Q-network takes the concatenation of the lower state and the scalar action as input. The minimum of Q_1 and Q_2 is used for target computa-

tion. Both networks are accompanied by exponential moving average target networks with rate $\tau = 0.005$.

Cost Q-network Q_c.. A single MLP with the same architecture as one branch of the twin Q-network estimates the expected cumulative cost (headway deviation) under the current policy:

Layer	**Shape**	**Activation**
Linear	$(8{+}K{+}1) \to 32$	ReLU
Linear	$32 \to 32$	ReLU
Linear	$32 \to 1$	—

The cost critic is trained with the same target-network mechanism and soft update rate as the reward critics. Its output enters the policy loss through the Lagrangian multiplier λ: the policy minimizes $\alpha \log \pi - Q + \lambda Q_c$.

Appendix D. TAP Derivation: From Synchronous to Asynchronous Cross-Advantage

This appendix formalizes the connection between the synchronous cross-advantage of **(author?)** [5] and the asynchronous Temporal Advantage Propagation (TAP) used in TransitDuet.

Appendix D.1. Synchronous Cross-Advantage (RoboDuet)

In the synchronous setting of **(author?)** [5], two agents—an upper-body controller and a lower-body locomotor—act at every time step t on a shared

state s_t. The cross-advantage augmented objectives are:

$$\mathcal{L}_{\text{upper}}^{\text{sync}} = -\mathbb{E}_t\big[(A_{\text{U}}(s_t, a_t^{\text{U}}) + \beta\, A_{\text{L}}(s_t, a_t^{\text{L}}))\, \nabla \log \pi_{\text{U}}(a_t^{\text{U}} \mid s_t)\big], \tag{D.1}$$

$$\mathcal{L}_{\text{lower}}^{\text{sync}} = -\mathbb{E}_t\big[(A_{\text{L}}(s_t, a_t^{\text{L}}) + \beta\, A_{\text{U}}(s_t, a_t^{\text{U}}))\, \nabla \log \pi_{\text{L}}(a_t^{\text{L}} \mid s_t)\big], \tag{D.2}$$

where $\beta \in [0, 1]$ controls coupling strength and both advantages are evaluated at the same state-action pair $(s_t, a_t^{\text{U}}, a_t^{\text{L}})$. This formulation is valid because the two policies share a common clock and state space.

Appendix D.2. The Asynchrony Challenge

In TransitDuet, the upper policy π_{U} and lower policy π_{L} operate on different timescales with different state-action spaces. Let the upper policy act at dispatch events $\{t_1^{\text{U}}, t_2^{\text{U}}, \ldots, t_N^{\text{U}}\}$, and the lower policy act at station arrivals $\{t_1^{\text{L}}, t_2^{\text{L}}, \ldots, t_M^{\text{L}}\}$, where $M \gg N$. These event sets are disjoint in general, and the state spaces differ: $s_{\text{U}} \in \mathbb{R}^5$ versus $s_{\text{L}} \in \mathbb{R}^{8+K}$. Thus, there is no single time step at which A_{U} and A_{L} are jointly defined on a shared state, and the direct sum in Equations (D.1) and (D.2) is undefined.

Appendix D.3. Trip-Lifecycle Alignment

We resolve this by introducing a *trip-lifecycle* alignment. Each upper action $a_{\text{U},i}$ at dispatch event t_i^{U} initiates a trip τ_i. During the lifecycle of trip τ_i, the lower policy makes a variable number of holding decisions $\{(s_j^{\text{L}}, a_j^{\text{L}}, r_j^{\text{L}})\}_{j \in \mathcal{J}(i)}$, where $\mathcal{J}(i)$ indexes all lower-level transitions belonging to trip i.

Definition 1 (Trip-aggregated lower signal)**.** *For trip i, the aggregated lower signal is:*

$$\bar{R}_{\text{L}}^{(i)} = \frac{1}{|\mathcal{J}(i)|} \sum_{j \in \mathcal{J}(i)} r_j^{\text{L}}. \tag{D.3}$$

Appendix D.4. TAP Augmented Objectives

Using the trip-lifecycle alignment, we define the TAP-augmented objectives:

$$\text{Upper:} \quad \tilde{R}_{\mathrm{U}}^{(i)} = R_{\mathrm{U}}^{(i)} + \beta\, \bar{R}_{\mathrm{L}}^{(i)}, \tag{D.4}$$

$$\text{Lower:} \quad \tilde{r}_j^{\mathrm{L}} = r_j^{\mathrm{L}} + \beta\, R_{\mathrm{U}}^{(\mathrm{trip}(j))}, \quad \forall\, j, \tag{D.5}$$

where $R_{\mathrm{U}}^{(i)}$ is the upper-level reward attributed to trip i (computed from the measurement vector at episode end), and $\mathrm{trip}(j)$ maps lower transition j to its parent trip.

Appendix D.5. Reduction to the Synchronous Case

Proposition 1. *If both policies share a common clock (i.e., each upper action triggers exactly one lower action at the same time step) and a common state space, then TAP reduces to the synchronous cross-advantage of* **(author?)** *[5].*

Proof. When $|\mathcal{J}(i)| = 1$ for all trips i, the aggregation in Equation (D.3) becomes $\bar{R}_{\mathrm{L}}^{(i)} = r_j^{\mathrm{L}}$ for the unique $j \in \mathcal{J}(i)$. The upper augmented return (Equation (D.4)) becomes $\tilde{R}_{\mathrm{U}}^{(i)} = R_{\mathrm{U}}^{(i)} + \beta\, r_j^{\mathrm{L}}$, which is the per-step sum $A_{\mathrm{U}} + \beta\, A_{\mathrm{L}}$ evaluated at the shared time step. Similarly, the lower augmentation (Equation (D.5)) reduces to $\tilde{r}_j = r_j^{\mathrm{L}} + \beta\, R_{\mathrm{U}}^{(i)}$, recovering $A_{\mathrm{L}} + \beta\, A_{\mathrm{U}}$. When the state spaces further coincide ($s_{\mathrm{U}} = s_{\mathrm{L}} = s_t$), this is exactly Equations (D.1) and (D.2). □

Appendix D.6. Policy-Gradient Unbiasedness

Proposition 2. *Under the standard assumptions of the policy gradient theorem (ergodicity and differentiability of π_{U}, π_{L}), the TAP-augmented policy*

gradient for the upper level,

$$g_{\mathrm{U}}^{TAP} = \frac{1}{N} \sum_{i=1}^{N} \tilde{R}_{\mathrm{U}}^{(i)} \, \nabla \log \pi_{\mathrm{U}}(a_{\mathrm{U},i} \mid s_{\mathrm{U},i}), \tag{D.6}$$

is an unbiased estimator of $\nabla_{\pi_{\mathrm{U}}} J(\pi_{\mathrm{U}})$ *when* $\beta = 0$*, and for* $\beta > 0$ *it is an unbiased estimator of* $\nabla_{\pi_{\mathrm{U}}}[J_{\mathrm{U}}(\pi_{\mathrm{U}}) + \beta \, J_{\mathrm{cross}}(\pi_{\mathrm{U}}, \pi_{\mathrm{L}})]$*, where* J_{cross} *is the expected trip-aggregated lower return induced by the upper policy's dispatch decisions.*

Proof sketch. The term $R_{\mathrm{U}}^{(i)} \, \nabla \log \pi_{\mathrm{U}}$ is the standard REINFORCE estimator for ∇J_{U}. The cross-term $\beta \, \bar{R}_{\mathrm{L}}^{(i)} \, \nabla \log \pi_{\mathrm{U}}$ has the form of a REINFORCE estimator where the "reward" for the upper policy at dispatch i is replaced by $\bar{R}_{\mathrm{L}}^{(i)}$, the average lower reward accumulated during the trip that dispatch i initiated. Conditional on π_{L} being fixed, $\bar{R}_{\mathrm{L}}^{(i)}$ is a function of the environment trajectory seeded by the upper action $a_{\mathrm{U},i}$, so the REINFORCE identity applies and $\mathbb{E}[\bar{R}_{\mathrm{L}}^{(i)} \, \nabla \log \pi_{\mathrm{U}}(a_{\mathrm{U},i} \mid s_{\mathrm{U},i})] = \nabla_{\pi_{\mathrm{U}}} \mathbb{E}[\bar{R}_{\mathrm{L}}^{(i)}]$. Linearity of expectation yields the result. □

An analogous argument holds for the lower-level augmentation: the injected term $\beta \, R_{\mathrm{U}}^{(\mathrm{trip}(j))}$ acts as a state-independent baseline shift (constant within a trip), which does not introduce bias in the lower policy gradient but can reduce variance by aligning lower-level credit assignment with upper-level performance.

Algorithm 2 TransitDuet Training Procedure

Require: Episode budget E, stage boundaries e_1, e_2, coupling limit $\beta_{\max}$, lower update frequency f_{L}, batch size B, fleet limit N_{fleet}

Initialize upper policy π_{U} with parameters ϕ
Initialize lower policy π_{L} with parameters ψ, twin critics $Q_{\omega_1}, Q_{\omega_2}$, cost critic Q^c_ξ, target networks $\bar{\omega}_1, \bar{\omega}_2, \bar{\xi}$
Initialize replay buffer $\mathcal{D} \leftarrow \emptyset$, Lagrange multiplier $\lambda \leftarrow 0$, entropy temperature α
Initialize weight vector $\boldsymbol{\theta} \leftarrow [-0.5,\ -0.3,\ -0.2]$
for episode $e = 1, \ldots, E$ **do**
  Compute $\beta(e)$ via eq. (23)
  if $e \leq e_1$ **then**
    Set $h^{\text{target}} \leftarrow 360\,\text{s}$ (fixed default timetable)
  else
    Sample $a_{\mathrm{U}} \sim \pi_{\mathrm{U}}(\cdot \mid s_{\mathrm{U}})$; extract $[H_{\text{peak}}, H_{\text{off}}, H_{\text{trans}}]$
  end if
  Reset simulation environment
  $\mathcal{T}_{\mathrm{U}} \leftarrow \emptyset$, $\mathcal{T}_{\mathrm{L}} \leftarrow \emptyset$, step $\leftarrow 0$
  while episode not done **do**
    if dispatch event triggered **then**
      Record $(s_{\mathrm{U},i}, a_{\mathrm{U},i})$ in $\mathcal{T}_{\mathrm{U}}$; select h^{target} based on time-of-day
    end if
    if station arrival for any bus b **then**
      Observe s_{L}; sample $a_{\mathrm{L}} \sim \pi_{\mathrm{L}}(\cdot \mid s_{\mathrm{L}})$; apply holding a_{L}
      Observe $r_{\mathrm{L}}, c_{\mathrm{L}}, s'_{\mathrm{L}}$; store $(s_{\mathrm{L}}, a_{\mathrm{L}}, r_{\mathrm{L}}, c_{\mathrm{L}}, s'_{\mathrm{L}})$ in $\mathcal{D}$ and $\mathcal{T}_{\mathrm{L}}$
      step $\leftarrow$ step $+1$
      if step mod $f_{\mathrm{L}} = 0$ **and** $|\mathcal{D}| \geq B$ **then**
        Sample batch of size B from $\mathcal{D}$
        Update $Q_{\omega_1}, Q_{\omega_2}$ via eq. (15)
        Update Q^c_ξ analogously
        Update π_{L} via eq. (17)

Table A.5: Complete hyperparameter specification.

Group	Parameter	Value	Description
Environment	`route_sigma`	1.5	Log-normal speed noise σ
	`max_agent_num`	25	Maximum concurrent buses
	`effective_trip_num`	264	Total trips per service day
Upper policy	`state_dim`	5	[hour, demand, fleet, prev_hw, unhealthy]
	`action_dim`	3	$[H_{\text{peak}}, H_{\text{off}}, H_{\text{trans}}]$
	`action_low`	$[180, 300, 240]$	Min headway per period (s)
	`action_high`	$[600, 1200, 900]$	Max headway per period (s)
	`hidden_dim`	64	MLP hidden layer width
	`lr`	10^{-4}	Policy learning rate (Adam)
	N_{fleet}	12	Fleet size hard constraint
Lower policy	`hidden_dim`	32	MLP hidden layer width
	`action_range`	60	Max holding time (s)
	c_{lim}	0.15	Lagrangian cost budget
	`batch_size`	2048	Replay buffer mini-batch
	`lr`	10^{-5}	Critic + policy learning rate
	λ_{lr}	10^{-3}	Lagrangian multiplier LR
	γ	0.99	Discount factor
	τ	0.005	Soft target update rate
	`auto_entropy`	true	Automatic entropy tuning
	$\alpha_{\max}$	0.3	Maximum entropy temperature
	`reward_scale`	10	Reward scaling factor
Coupling (TAP)	β_{warmup}	50 episodes	Stage 1: only lower trains
	β_{ramp}	100 episodes	Stage 2: β ramps $0 \to \beta_{\max}$
	$\beta_{\max}$	0.5	Maximum cross-advantage weight
	$\eta_{\boldsymbol{\theta}}$	0.01	ApproPO $\boldsymbol{\theta}$-OGD step size
Training	`total_episodes`	300	Total training episodes
	`save_freq`	50	Checkpoint frequency (episodes)
	`replay_buffer_size`	10^6	Replay buffer capacity